Educated to Be Mothers? School Reform and Demographic Backlash*

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November 7, 2025

Abstract

Social norms play a critical role in society. This paper studies how difficult it is to manipulate social norms and whether efforts to do so may result in a backlash. We study the 1945 reform of primary education implemented under Franco's regime in Spain, which promoted nationalist-religious values and emphasized women's domestic roles. Exploiting plausibly exogenous variation in exposure between birth cohorts, we show that women exposed to the reform have fewer children and tend to reject the regime's promoted gender norms. These findings highlight the unintended long-term consequences that state-led efforts to engineer social norms can generate.

JEL Classification: I21, J13, N44

Keywords: Education, Fertility

*We thank Noemi Fachetti for excellent research assistance. We also thank Arnstein Aassve, Davide Cipullo, Gianmarco Daniele, Federico Masera, Mounu Prem and participants of the Summer Crime Workshop 2024 at Bocconi University, the 4th Winter Symposium in Economics and Finance at the Catholic University of Milan, the departmental seminar at UNSW Sydney, the AXA Gender Lab Seminar Series at Bocconi University, the ALP-POP Conference (DONDENA, Bocconi University), the 4th Junior Economists Meeting at University of Milan and the Economic History Seminar at Universidad Carlos III of Madrid for invaluable comments and suggestions.

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1 Introduction

During the past half century, fertility rates have dropped dramatically worldwide (Goldin, 2025). The global total fertility rate has decreased from about 5 children per woman in 1960 to about 2.3 today, and a growing number of countries are now below the replacement level of 2.1 births per woman (Rindfuss et al., 2010; Hart et al., 2024). This has raised concerns about population aging, shrinking workforces, and associated economic challenges (Hart et al., 2024). Many governments have adopted pro-natalist policies aimed at encouraging childbearing and, by the mid-2010s, about 66% of European countries and 40% of Asian countries had policies to increase fertility (Hart et al., 2024). These policies take various forms, including direct financial incentives (baby bonuses, child allowances, tax credits), expanded parental leave, subsidized childcare, housing benefits for families, and other measures to reduce the cost of childbearing.

Although evidence suggests that lowering the burden of child care for mothers should be particularly effective in boosting fertility (Doepke and Kindermann, 2019; Sakamoto and Minami, 2024), a growing share of policymakers have begun to advocate for a return to traditional gender roles, linking declining fertility with changes in gender norms, family formation patterns, and work–life preferences. In the UK, prominent figures have called for broad cultural changes to encourage childbearing and to recenter family life within national identity.[1] Elsewhere in Europe, political leaders have positioned demographic recovery as contingent on restoring 'traditional family values' and prioritizing motherhood within national policy agendas.[2] In the US, pro-natalist arguments argue that declining birth rates reflect the erosion of stable family structures and cultural norms that once supported childrearing.[3] From this perspective, strengthening conventional gender roles and family arrangements is presented as a necessary foundation for demographic renewal.

This paper evaluates the validity of this claim by empirically examining a primary school reform that attempted to boost fertility by stressing traditional gender roles. The reform was enacted by the dictatorial regime of General Francisco Franco in Spain in 1945. It outlined the key characteristics of primary education, emphasizing religious instruction, strong national spirit, mandatory use of the Spanish language, and a specialized curriculum for girls focused on home life, crafts, and domestic industries. It mandated the separation of sexes and established strict guidelines for teaching practices, carefully regulating time management and the structure of the school day.[4]

A key goal of the reform was to educate women to become "good wives" and "good mothers",

[1] See, e.g., Cates (2023), Farage (2025a), and Farage (2025b).

[2] See, e.g., Erdogan (2016), Meloni (2023a), Meloni (2023b), and Orbán (2024).

[3] See, e.g., Vance (2025a) and Vance (2025b).

[4] More details about primary school are provided in the next section.

as "it would be the most appropriate way for them to contribute to society" (Toboso Sánchez, 2009, p. 213). As such, this initiative was not an isolated case. Historical examples of autocratic regimes trying to interfere in the private lives and fertility decisions of their citizens through propaganda and/or indoctrination include the Nazi Lebensborn program to encourage fertility among "racially valuable" German women (US Holocaust Memorial Museum), female social organizations under the Pinochet regime in Chile (F. González et al., 2024), Franco's National Birthrate Awards and the award of Medals of motherhood in the Soviet Union.[5]

To uncover the causal effect of Franco's primary school reform on fertility behavior, the analysis uses both nationally representative survey data and census microdata at the individual level. We exploit quasi-exogenous variation in exposure to Franco's 1945 primary school reform originating from the interaction between the birth cohorts of individuals, the date of implementation of the reform, and the official age of entry and exit from primary education. In particular, we compare fertility choices between women born in the 1920s, who completed primary school before the reform, in the 1930s, who were partially exposed to the reform, and from 1939 to 1950, who attended the entire primary education cycle after the reform. Therefore, our intention-to-treat (ITT) measure distinguishes between partial and full exposure to the reform. Our estimates show that women more exposed to the reform - solely due to the timing of school entry - were equally likely to become mothers, but they had significantly fewer children on average. Thus, the reform seemingly influenced fertility decisions primarily on the intensive margin. Additional analysis using the Spanish census microdata and a set of placebo tests retrieve results that further validate these conclusions.

Additional tests do not support a set of hypothesized mechanisms, including differential participation in the labor market, educational achievement, and political leaning. In contrast, our results provide at least suggestive evidence that the decline in fertility among treated cohorts was accompanied by a change in beliefs plausibly shaped by the reform, particularly regarding gender roles, motherhood, and the family. These results are consistent with the presence of a backlash effect, in which the regime's attempt to engineer traditional values ultimately undermined the very same norms it sought to establish.

Franco's primary school reform was a bundled treatment that generated a backlash. Data limitations do not allow us to quantify the extent to which this was caused by mere exposure to the reform versus other factors, such as the different curricula, the division of classes, or the different school materials between males and females. However, to investigate whether these individual-level effects aggregated into broader demographic shifts, the final part of the paper turns to historical census data and estimates a series of difference-in-differences specifications at the cohort–locality level. Here, the treated group is defined as the female population of primary

[5]See, e.g., the *Medal of Motherhood (Central Asia)*, available at: https://www.cabinet.ox.ac.uk/medal-motherhood-central-asia-0

school age as a fraction of total population in each locality, measured from the 1940 census. Thus, we compare fertility trends between areas that differed in the share of school-age girls at the onset of the reform, a demographic factor that plausibly shaped the intensity of exposure to the reform between localities, and subsequently influenced fertility outcomes as these cohorts reached reproductive age. This analysis reveals a significant decline in fertility rates in areas with a greater share of the population exposed to the reform, while marriage rates remained unaffected. A variety of robustness checks confirm this conclusion and placebo tests either using the share of school-age boys in 1940 or the share of women aged 15-24 in 1940 yield null effects, mitigating concerns that we might be capturing pre-existing demographic trends.

Our paper contributes to four different strands of literature. First, it contributes to a large body of work on fertility and family policies. These include transfers, such as one-time "baby bonuses", recurring child allowances, and tax deductions for families with children. By lowering the monetary cost of an additional child, transfers have the potential to increase fertility (Milligan, 2005; Cohen et al., 2013; L. González, 2013; Slonimczyk and Yurko, 2014; Malak et al., 2019; Tudor, 2020; L. González and Trommlerová, 2023; Kim et al., 2023). Common family policies also include parental leave policies that guarantee job-protected leave from work for parents after a child is born, often with some wage replacement. The evidence shows mixed but generally positive effects of such reforms on fertility (Lalive and Zweimüller, 2009; Dahl et al., 2016; Cannonier, 2014; Golightly and Meyerhofer, 2022; Bailey et al., 2025), with clearer impacts linked to more substantial leave reforms or to contexts with previously very limited support (see, e.g., Raute, 2019).[6] Family policies that have been studied also include childcare support policies, which appear to be particularly effective, as evidence from multiple different countries has consistently shown positive fertility effects (Rindfuss et al., 2010; Havnes and Mogstad, 2011; Mörk et al., 2013; Bauernschuster et al., 2016; Wood and Neels, 2019). Other measures that have been tried or proposed to encourage childbearing include housing policy (Clark, 2012; Fazio et al., 2024), tax incentives (Parent and Wang, 2007; Cohen et al., 2013; Laroque and Salanié, 2014), health and family planning (Abramowitz, 2018; Gartner et al., 2022), and broader socio-cultural campaigns (Zeng and Hesketh, 2016; Lin et al., 2024). However, no studies to date have empirically examined whether an educational reform can work to stimulate fertility in future generations. Our results rectify this omission and show that Francoist primary school reform backfired and resulted in reduced fertility. This conclusion is of primary importance for a policy maker interested in combating the modern demographic winter.

Second, this article is related to existing research on gender norms and how they affect the lives of women in many dimensions (Bertrand et al., 2015; Fortin, 2015; Goussé et al., 2017; Andresen and Nix, 2019; Grewenig et al., 2020; Olivetti et al., 2020; Cortés and Pan, 2023;

[6] Recent studies also find that giving fathers parental leave (paternity leave quotas or shareable leave) causes little change in fertility (Lee et al., 2009; Cools et al., 2015; Farré and L. González, 2019; Duvander et al., 2020).

Boelmann et al., 2025). Gender norms play a crucial role in defining gender inequality (Kleven, Landais, Posch, et al., 2019; Bertrand, 2020; Ashraf et al., 2022; Bursztyn et al., 2023; Farré, Felfe, et al., 2023), as they generally assign women the role of primary caregivers within the family and restrict their options in the labor market after childbirth (Bertrand et al., 2015; Fortin, 2015; Goussé et al., 2017; Andresen and Nix, 2019; Grewenig et al., 2020; Olivetti et al., 2020; Cortés and Pan, 2023; Boelmann et al., 2025). Although modern gaps can be traced at least in part to parenthood (Fernández et al., 2004; Alesina, Giuliano, et al., 2013; Farré and Vella, 2013), which has the potential to hold back women's careers relative to men (Daniel et al., 2013; Angelov et al., 2016; Kleven, Landais, and Søgaard, 2019; Bertrand, 2020; Cortés and Pan, 2023) and perpetuate gender imbalances (Albanesi et al., 2023; Kleven, Landais, Posch, et al., 2024), public policy has the potential to effectively promote gender equality in the long term (Hara and Rodriguez-Planas, 2021; Farré, Felfe, et al., 2023). Our paper contributes to this debate by showing that opposite policies, i.e., that stress traditional gender roles, are unlikely to achieve their intended purpose and may reduce fertility. This finding echoes Goldin (2025)'s conclusion that some women will not have children and others will delay when the probability is low that men will abandon traditions.

Third, our study is related to existing research on the use of education and propaganda by authoritarian regimes (Waldinger, 2010; Waldinger, 2012; Adena et al., 2015; Voigtländer and Voth, 2015; Fuchs-Schündeln and Masella, 2016; Cantoni et al., 2017; Homola et al., 2020; F. González et al., 2024; Meriläinen and Mitrunen, 2025; Grebol and Navarro, 2025). In authoritarian regimes, as in modern democracies, the content and quality of education have lasting effects on people's beliefs. Germans who grew up under the Nazi regime had access to significantly lower quality of education (Waldinger, 2010; Waldinger, 2012) and exhibit higher levels of anti-Semitism than those born before or after that period (Voigtländer and Voth, 2015). East Germans who were exposed to communism are more in favor of state intervention than West Germans (Alesina and Fuchs-Schündeln, 2007) and prolonged exposure to socialist education reduced their chances of obtaining a college degree (Fuchs-Schündeln and Masella, 2016). In Spain, Catalan teaching was effective in increasing Catalan feelings (Clots-Figueras and Masella, 2013), while in China modifying the school curriculum was effective in changing young people's views of China's governance and free trade (Cantoni et al., 2017). In contrast, language restrictions in elementary schools in the US after World War I instigated a backlash, increasing the sense of cultural identity among the minority, reducing integration and identification with the host country, and showing that propaganda may not generate the intended effects (Fouka, 2020). We contribute to this literature by shedding new light on how state-led value engineering—via authoritarian educational reform—can leave a lasting imprint on childbearing decisions.

Finally, our analysis contributes to existing research on the formation of cultural beliefs, the marriage market, and fertility (Chong and La Ferrara, 2009; La Ferrara et al., 2012). Although

existing studies show that *novelas*, soap operas and TV series can influence marital and fertility decisions, our findings highlight that school reforms stressing traditional gender norms are ineffective pro-natalist policy tools.

The remainder of the paper is structured as follows. Section 2 reviews the historical background of Spain under the Francoist regime. Section 3 shows our data sources, Section 4 describes the empirical analysis, and Section 5 presents the results. Finally, Section 6 concludes.

2 Historical background

In 1936, a military uprising led by Francisco Franco aimed at overthrowing the Republican government unleashed the Spanish Civil War. The conflict between Republicans and Nationalists, which ended in 1939 with Franco's victory, resulted in approximately 500,000 fatalities and around 450,000 exiled people (Castro Oury, 1993; Preston, 2011, p. 608). Following this victory, Franco ruled over Spain from 1939 to 1975 as a dictator.

The Francoist regime repeatedly expressed its determination to implement demographic policies, considering fertility to be one of the key concerns of the state (Nash, 1991, p. 160). Ramón Serrano Suñer- president of the *Falange Española Tradicionalista y de las Juntas de Ofensiva Nacional-Sindicalista, FET-JONS*, i.e., the sole legal political party in Francoist Spain - pointed out the necessity of a *"serious demographic policy to accomplish the ideas of the Falangist State"*. In the words of Ramiro Ledesma, founder of the political party *JONS*: *"we can only seriously speak of Spain's superiority when it becomes attainable through a population that has at least doubled" (Bosch Marín, 1941, p. 33)*. In fact, maternity was described as an *"ordinary way of life to reach perfection"* (Martínez Medrano, 1959, p. 31) and represented the feminine goal: *"the feminine profession should be oriented towards family life: It belongs to the very essence of a woman to be always and everywhere a mother, to devote and sacrifice herself, and to care with care for others"* (Jaén Sánchez, 1959, p. 21).

As part of Franco's agenda to boost fertility rates in post-civil war Spain, the antiabortion and anticontraception law was soon approved in 1941 (Spanish Official Bulletin, 1941, p. 768-770).[7] In 1945, the first Primary Education Law of the Francoist regime was also enacted. The reform established that the specific education of women as wives, mothers, and educators of their children should begin in primary school (Rabazas Romero and Ramos Zamora, 2006, p. 46 and Szczygieł, 2014, p. 295-296). The reform was openly endorsed by the primary education inspector, Francisca Bohigas, who claimed that *"the girl was born to stay at home"* (p. 10,

[7]This law does not contaminate our analysis because it applied uniformly across Spain and to all cohorts in our sample, and any fertility effect it may have had will be absorbed by the year- and province-by-year fixed effects included in our analysis, as described in more detail in Section 4. Our research design ensures this reform cannot plausibly drive the cohort-specific patterns that emerge only after the 1945 education reform.

Bohigas, 1941), and it was backed by the Catholic Church and the Women's Section of the Falange (Toboso Sánchez, 2009, p. 212).

Primary school consisted of three phases: elementary education (6-10 years old), improvement (10-12 years old), and vocational initiation (12-15 years old).[8] The law defined the main characteristics of primary education: a different curriculum for girls that prepares them for home life, crafts, and domestic industries, among others because they had to prepare to become mothers and wives (Rabazas Romero and Ramos Zamora, 2006, p. 47; Francés Díez, 2013, p. 226). In addition, the law mandated the inclusion of religious education, emphasized the promotion of strong national identity, and enforced the use of the Spanish language. It promoted values that differed from the previous education law *(Royal Decree of May 6, 1931)*, which had established a secular coeducational system with bilingual instruction (Morente Valero, 1997, p. 59-61). The role of teachers was subordinated to the supreme interests of the nation, with the responsibility of contributing to the formation of girls as future mothers.

The Female Section also contributed to the propaganda efforts of the regime through the journal *Consigna*, published from 1940 to 1977 and containing pedagogical guidelines for female teachers, mothers, or future mothers (Rabazas Romero and Ramos Zamora, 2006, p. 49–50):[9]

> *"From girlhood, a woman is feminine, and her differentiation from boys is evident even in the smallest details and games. The care invested in her education will later be reflected in the home, where the girl, now a woman, must shape all those around her: husband, children, [...] And isn't it a serious responsibility of a teacher—or an educational system—not to have fully nurtured the seed that was planted? [...] If we enable our women to become complete women, we will have eliminated many of the unhappy and uncomfortable households that exist in our society today. Because the mother woman will raise her daughters just as we have raised her [...]. This is a call to the conscience and responsibility of every female teacher." (Consigna, 1958, p. 30–32,)*

The regime issued precise guidelines on methodological practices and strictly regulated time usage during the school day. In addition, it mandated the separation of sexes in primary education and kindergarten, with co-educational schools (in cases of very low enrollment) to be run by female teachers. The law also established the compulsory subjects to be taught in school: core subjects (such as writing and arithmetic), formative subjects (such as mathematics, religion and national history), and complementary subjects (such as feminine tasks, including domestic economy and family life education; see Toboso Sánchez, 2009, p. 215). Moreover, some schools

[8]School attendance was compulsory from 6 to 12. Penalties were imposed on those who required children to work. For the first time, a school record book and a primary education certificate were established as documents attesting to the compulsory nature of education.

[9]The Female Section (*Sección Femenina*) was founded in 1934 and led by Pilar Primo de Rivera, sister of José Antonio Primo de Rivera, founder of Falange. Starting in 1937, a Social Service, organized by the *Sección Femenina*, was introduced for women aged 17-35 who aspired to occupy positions in the administration or obtain university degrees (Gobierno del Estado, 1937). However, also due to the endogenous self-selection of women in this program, the *Sección Femenina* is not the subject of our study.

organized professional initiation courses for children aged 12-15 years old.[10] School material, such as books, had to be approved by the Ministry of National Education and the ecclesiastical hierarchy (González Pérez, 2009, p. 97).[11]

Numerous examples of the different reading books used to educate boys and girls during the Francoist regime document the indoctrination effort perpetrated by the regime. Girls received books that narrated the birth of the protagonists (though without referencing sex, instead attributing it to God or the stork), stories where young girls were responsible for entire families, or stories emphasizing the important role of women as mothers (Mahamud, 2016), as in this fragment of the book titled *"Read me, girls"* (see also Figure 1):

> *"Little girls, budding women:*
> *I offer you one teaching: May you always feel in your soul the sacred fear of God, may you always be sweet little mothers [...], eyes to ease pain, little hands to heal suffering. [...] you will always be the sweetest, the most beautiful, the most delicate in the world." (Torres, 1943, p. 158)*

Figure 1: Cover and excerpt from *Leedme, niñas. Segunda parte* (Torres, 1943).

[10]In particular, the law differentiated "domestic-type female facilities where students, according to their gender, would carry out appropriate practices to develop work habits, prepare for professional life, and their future family roles." The distinction between arithmetic and mathematics reflected the regime's view of the former as a practical skill and the latter as part of intellectual and moral training.

[11]See the Spanish Official Bulletin (1945, p. 385-394).

The volume *"Guirnaldas de la Historia"*, intended for girls, is another example that describes procreation as the ultimate goal to contribute to the nation (Criado and De los Reyes Leoz, 2019, page 292). This vision of womanhood was also consistently reinforced in other materials:

> *"We can prepare as good mothers those that are still girls" (Paya Ibars, 1959, p. 7–11)*

> *"The boy will set his eyes on the world; the girl, on the home" (Toboso Sánchez, 2009, p. 215)*

The historical evidence presented here is of course purely anecdotal, but it exemplifies how interfering in women's fertility decisions was a central objective of the 1945 Francoist primary education reform.

3 Data

To examine the impact of the 1945 Francoist primary education reform on fertility at the individual level, the empirical analysis draws on both the *Centro de Investigaciones Sociológicas* (CIS) barometer (i.e., Center of Sociological Investigations) and specific thematic surveys as primary data sources. CIS surveys are nationally representative and have been administered at irregular intervals since 1979. In addition to collecting standard sociodemographic information, these surveys elicit attitudes, values, and beliefs of the respondents on a wide range of topics, including politics, religion, gender roles within the family and society, and views on child rearing. From the 1979 and 1990-1992 waves, we extracted information on the total number of children ever born to each female respondent, which we used to construct our main individual-level outcome variables - both at the extensive and intensive margins. We also obtained information on each female respondent's year of birth, which allows us to measure exposure to Franco's school reform, and location of residence at the time of the survey.[12] From the 1979 and 1986–1988 waves, we further collected data on the labor force participation of women (i.e., if women are employed, unemployed or housewives), their educational attainment, and whether they voted for right-wing parties in the last elections.[13] To explore potential mechanisms—particularly the hypothesis of a cultural backlash triggered by Franco's school reform—we turned to the 1990 survey, which includes several questions probing female respondents' beliefs about women's role in the family, motherhood, female labor market participation, and alignment with Catholic

[12] CIS data provide the exact municipality of residence only for localities with more than 100,000 inhabitants; thus, we classified localities into provincial capitals versus the rest of the province.

[13] Information on the labor force participation and women's educational attainment is available from the 1979 and 1986–1988 waves, while voting behavior is available only from the 1986-1988 waves.

values.[14]

To further validate the findings obtained from the CIS surveys, we complemented our analysis with two additional sources of individual-level data: the Spanish census microdata produced by the *Instituto Nacional de Estadística* (INE), i.e., the National Institute of Statistics of Spain, and the European Social Survey (ESS). Census microdata allow us to improve the precision of our estimates due to their administrative nature and substantially larger sample sizes. However, they display limited information, since they lack the rich attitudinal content of CIS surveys—such as beliefs, values, and social norms—and do not include certain demographic details, including childbearing information for men. As such, our analysis based on census data is limited to testing whether our main result, the effect of exposure to Franco's primary school reform on women' fertility behavior, can be replicated in administrative data. Moreover, using the ESS data, we conducted a cross-country placebo test using a state comparable to Spain in several key aspects but not exposed to the Francoist reform, namely Portugal.[15] This exercise allows us to assess whether the pattern observed in Spain may be driven by a broader cohort effect rather than the reform itself.

In both the Spanish census microdata and the ESS data, we built our dependent variables following the same logic as in the CIS-based analysis. From the census, we used the 1991 and 2011 waves,[16] while from the ESS we used the 2006 and 2018 waves.[17] As in the census data, we used information on the sex of the respondents and the municipality of residence at the time of the survey,[18] and measured exposure to Franco's school reform based on the year of birth of the respondent, which is consistently reported in both data sources.

To complement our baseline individual-level analysis and assess if the effects of Franco's primary school reform also appear at the aggregate level, we compiled a panel dataset spanning nearly five decades (1930–1979). The units of observation are the provincial capitals and the rest of their respective provinces, based on data availability across sources. Our primary outcome variable here is a measure of fertility constructed from the annual Natural Movement of the Population reports published by INE. These reports provide disaggregated birth statistics for each provincial capital and for the rest of the province, allowing for long-term spatial and

[14]More specifically, we use the answer to questions such as: *Of the following types of families, which one best matches your ideal family? To what extent do you think a man should share household chores? Is sharing household chores very important to you? Is having children very important for achieving the stability of the couple? Do you believe that being a mother is an experience one must have in order to truly feel like a woman?* Appendix Table A2 shows the complete list of questions.

[15]Portugal's geographic proximity, strong Catholic tradition, and the presence of an authoritarian regime during the same historical period make Portugal an ideal comparison state for this analysis.

[16]The 1991 census is the earliest wave for which individual-level microdata are publicly available. We exclude the 2001 census, as it does not contain fertility information.

[17]These are the only waves containing information about childbearing decisions.

[18]Unfortunately, municipality of residence is not available in the ESS.

temporal comparisons.

Ideally, aggregate fertility would best be captured by the Total Fertility Rate (TFR), a weighted average of age-specific fertility rates for women of childbearing age (aged 15 to 44). The TFR allows meaningful comparisons both within and across observational units. However, the Natural Movement data report only the absolute number of live births per unit and year, without disaggregating it by maternal age. To make this crude birth count comparable across time and space, we adjusted for the size of the relevant at-risk population. Specifically, we controlled for the number of women of reproductive age (aged 15 to 44) in each observation unit and decade, as reported in the Decennial Historical Censuses published by INE from 1930 to 1970.[19]

From the same INE historical censuses, we also collected data on the size of the female population aged 0 to 14 in 1940. This cohort serves as our proxy for exposure to Franco's school reform: being the first to be directly affected by the policy when it came into force in 1945, we expect its relative size in each observation unit to be predictive of the possible impact of the reform on future fertility patterns at the unit level as this cohort enters reproductive age in the following years. Figure B1 displays the spatial distribution of our aggregated measure of exposure to treatment. The intensity of exposure appears relatively homogeneous between the northern and southern regions of Spain. Some heterogeneity appears along the coastal-inland axis, as inland regions tend to exhibit higher levels of exposure compared to coastal areas. This spatial pattern provides reassurance that our empirical strategy is not confounded by broad North–South gradients, although finer geographic variation may still play a role in shaping the estimated effects.[20] Finally, we created two additional variables for our analysis: from the Natural Movement reports, we gathered annual data on the number of marriages per observation unit; from historical censuses, we gathered data on the size of the male population aged 0–14 in 1940 and the size of the female and male population aged 15-24 in 1940.[21]

4 Identification

To estimate the causal effect of the Francoist primary school reform introduced in 1945, our main analysis exploits the variation in exposure driven by the interaction between the year of birth of individuals, the timing of the reform, and the official age of entry and exit from school at the time of the reform.[22] In particular, we compare fertility choices between three

[19]To account for potential shifts in the age structure of the reproductive population, we further disaggregated this group into three subcategories—ages 15–24, 25–34, and 35–44—and included separate controls for each subgroup in our regressions.

[20]Our regression models account for this heterogeneity through the inclusion of fixed effects at the unit level.

[21]Appendix Table A1 displays summary statistics for the main variables, while Appendix Table A2 provides a list of the variables included in the analysis and their brief descriptions.

[22]As Section 2 describes, primary school in Spain spanned the ages 6 to 15. Before the Francoist reform, compulsory schooling roughly corresponded to the age range 6–14 (Instrucción Pública, 1923). After the reform,

groups of individuals: (i) those born between 1920 and 1929, who completed their entire primary education before the reform; (ii) those born between 1930 and 1938, who were partially exposed to the reformed curriculum; and (iii) those born between 1939 and 1950, who completed their full primary education under the post-reform system.[23] Specifically, we estimate the following specification on the sample of women interviewed in the CIS barometer surveys:

$$y_{it} = \beta_0 + \beta_1 \mathit{1930/1938}_i + \beta_2 \mathit{1939/1950}_i + Year_t + Residence_i + AgeClass_i + \epsilon_{it} \qquad (1)$$

where the dependent variable is either a dummy variable equal to 1 if woman i had at least one child by year t (and 0 otherwise) or a count variable measuring the total number of children of woman i by year t. The dummy variable *1930/1938* is equal to 1 if woman i was born between 1930 and 1938; the associated coefficient β_1 captures the effect of partial exposure to the reform on fertility outcomes, relative to the omitted category of women born between 1920 and 1929. Similarly, the dummy variable *1939/1950* equals 1 if woman i was born between 1939 and 1950, and the coefficient β_2 estimates the effect of full exposure to the reform on fertility outcomes.

The use of multiple survey waves allows us to disentangle the effect of the reform, which is inherently cohort-based, from any time-specific shocks by including year fixed effects ($year_t$). Although we cannot include fixed effects for the exact age of the respondents, we group the individuals into ten-year age brackets at the time of the survey and include fixed effects for these age groups in the regression ($AgeClass_i$).[24] We also control for $Residence_i$, a set of fixed effects that capture the respondent's place of residence at the time of the interview, classified as the provincial capital or the rest of the province. In our most comprehensive specification, we further allow for nonparametric trends that vary jointly by age class and place of residence by including the full set of interactions between the three groups of fixed effects. Standard errors are heteroskedasticity-robust and clustered at the level of birth cohort-survey year.[25]

To test the validity of the parallel trends assumption and to gain further insight into the dynamic effects of exposure to the reform in the subsequent years, we also estimate the event study specification in Equation 2 below:

compulsory education ended formally at age 12, although students could stay in school up to age 15 on a voluntary basis. This was often due to grade repetition, which was relatively common at the time especially in rural areas where illiteracy was widespread. Others who remained enrolled after age 12 had often failed the entrance exam to higher secondary school (*bachillerato*), had no access to vocational education, or wished to receive complementary instruction. To ensure consistency in the definition of primary schooling between the pre- and post-reform periods, we define primary education as spanning ages 6 to 15, and assign exposure accordingly.

[23] To maximize comparability, we restrict the last group to individuals born in 1950.

[24] Specifically, we define the following age groups: under 20, 20–29, 30–39, and so forth.

[25] Clustering by both birth cohort and survey year properly accounts for the correlation among respondents who share both the timing of exposure and the survey wave in which they were interviewed. However, Section 5.1 also presents robustness tests in which we use cohort-level clusters and wild cluster bootstrap to account for the small number of clusters, or two-way clusters at the cohort and survey level.

$$y_{it} = \alpha + \sum_{\substack{c=1920 \\ c \neq 1929}}^{1950} \beta_c \cdot \mathbf{1}\{\text{BirthYear}_i = c\} + Year_t + Residence_i + AgeClass_i + \epsilon_{it} \tag{2}$$

In this analysis, rather than grouping individuals into broader birth cohorts as described above, we include a set of indicator variables for each single-year birth cohort. The reference category is the latest cohort to have completed the entirety of primary school before the Francoist reform, namely those born in 1929.

5 Results

Table 1 presents our baseline estimates of the impact of Franco's primary education reform in 1945 on fertility outcomes. Columns 1–4 present estimates for our binary measure of fertility that captures whether woman i in Equations 1 and 2 had a child or not by year t (i.e., the extensive margin of fertility), while columns 5–8 present estimates for our continuous measure of fertility that captures the count of children of woman i in Equations 1 and 2 by year t (i.e., both the extensive and intensive margins of fertility). Columns 1 and 5 include only year fixed effects; Columns 2 and 6 add fixed effects for the place of residence; Columns 3 and 7 additionally include 10-year age group fixed effects; Columns 4 and 8 incorporate the full set of interactions among these three groups of fixed effects. As shown in Table 1, we find no statistically significant effect of the reform at the extensive margin: women who were partially or fully exposed to the reform appear equally likely to have had a child compared to the control group. However, when considering both the extensive and intensive margin, we find a statistically significant and negative effect limited to fully treated cohorts. In particular, women who completed their primary education under the reform exhibit a lower average number of children compared to untreated cohorts. In the most saturated specification (column 8), the reform appears associated with a reduction of approximately 0.34 children, equivalent to a 12% decrease relative to the average number of children among untreated women. In other words, the reform did not significantly influence the likelihood of becoming a mother, but it did affect the size of the family, that is, the number of children that women had on average. It is important to note that these estimates represent the average effect for women fully exposed to the reform. As we show in the event study analysis below, the decline in fertility begins to emerge among the early 1930s cohorts, which were partially exposed to the reform.

Using the specification in column 8 of Table 1 as a reference, Figure 2 presents the corresponding event study estimates. In Figure 2, the coefficients for the untreated cohorts are generally not statistically different from zero. Although some estimates are noisy and fluctuate around zero, there is no evidence of a statistically significant pretrend. In contrast, the negative effect of Franco's school reform begins to emerge from the 1932 cohort, who were 13 years old in 1945

Table 1: The impact of Francoist primary school reform on fertility

	(1) Kids	(2) Kids	(3) Kids	(4) Kids	(5) N. kids	(6) N. kids	(7) N. kids	(8) N. kids
1930/1938	0.012	0.010	-0.014	-0.006	-0.079	-0.061	-0.056	-0.177
	(0.013)	(0.013)	(0.020)	(0.019)	(0.069)	(0.067)	(0.106)	(0.131)
1939/1950	0.017	0.013	-0.004	0.002	-0.370***	-0.401***	-0.260**	-0.344**
	(0.010)	(0.009)	(0.025)	(0.025)	(0.065)	(0.063)	(0.128)	(0.149)
Observations	4,157	4,157	4,157	3,944	4,153	4,153	4,153	3,940
R-squared	0.001	0.023	0.025	0.175	0.015	0.068	0.069	0.209
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Residence FE	NO	YES	YES	YES	NO	YES	YES	YES
Age group FE	NO	NO	YES	YES	NO	NO	YES	YES
Year X Age group X Residence FE	NO	NO	NO	YES	NO	NO	NO	YES
Base value	.89	.89	.89	.89	2.84	2.84	2.84	2.84

Dependent variable: *Ever had a child* (Columns 1–4), *Number of children ever had* (Columns 5–8). All estimates are based on the 1979 and 1990–1992 waves of the CIS barometer surveys. *1930/1938* is a dummy equal to 1 for women born between 1930 and 1938; *1939/1950* is a dummy equal to 1 for women born between 1939 and 1950. The omitted category comprises women born between 1920 and 1929. Columns 1 and 5 include year fixed effects; Columns 2 and 6 add fixed effects for place of residence (provincial capital vs. rest of province); Columns 3 and 7 further include 10-year age group fixed effects; Columns 4 and 8 include the full interaction between year, residence, and age group fixed effects. Standard errors are robust to heteroskedasticity and clustered at the birth cohort-survey year level. The final row reports the mean of the dependent variable for the reference group (i.e., women born in 1920–1929). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

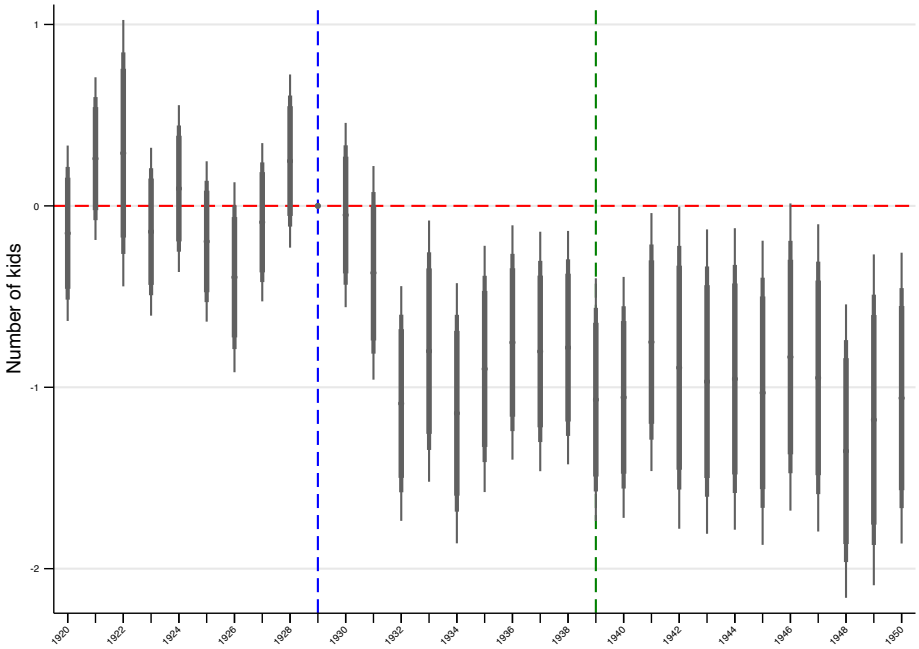
and spent three years under the reform. When examining post-treatment dynamics, the negative effect appears fairly stable in magnitude across cohorts, regardless of whether they were partially or fully exposed to the reform. The only exception are the 1930-1932 cohorts, who display a gradually-increasing estimated effect. As discussed in Section 4, these cohorts were exposed to the reform only during the non-compulsory years of schooling (ages 13–15). As a result, only a subset of women in these cohorts received treatment, in all likelihood with a greater fraction for women from the 1932 cohort, who were 13 years old in 1945, and a smaller fraction from the 1930 cohort, who were already 15 in 1945. The results in Figure 2 for the 1930-1932 cohorts appear sensible, as they show that the growing share of treated women over time among these cohorts resulted in a gradually-increasing estimated effect. Taken together, the results in Table 1 and Figure 2 indicate that Franco's 1945 primary school reform significantly reduced fertility among both partially and fully exposed women once they reached their reproductive age.

5.1 Robustness

As a first robustness test, Table B3 re-estimates the specifications shown in columns 4 and 8 of Table 1 using two alternative clustering strategies. Specifically, in columns 1 and 3, standard errors are clustered at the birth cohort level. Given the small number of clusters (i.e., only 31), we also report p-values based on the wild cluster bootstrap. Columns 2 and 4 instead adopt a two-way clustering approach at the birth cohort and survey year levels. In both cases, the results remain highly consistent with the baseline estimates presented in Table 1.

To further assess the robustness of our findings, we performed two additional exercises. First, we ran a validation test by reestimating Equation 1 on an alternative and substantially larger

Figure 2: The impact of Francoist primary school reform on fertility: Event Study



Note: The figure shows the point estimates and the confidence intervals at 90% (i.e. least-wide spikes) and at 95% (i.e. widest spikes) for each birth cohort of women in Equation 2 and for the specification presented in column 8 of Table 1. The dependent variable is the number of children of a woman. The blue dashed line indicates the reference birth cohort (i.e., the latest cohort that completed primary school before Franco's reform, who are women born in 1929), while the green dashed line indicates the first birth cohort fully affected by the reform, (i.e., the earliest cohort that completed the entire primary school cycle under the reform, who are women born in 1939).

dataset, the Spanish population census microdata collected by INE. Second, we performed a placebo test on a sample of individuals from Portugal using data from the European Social Survey (ESS).

Table B4 reports the estimates of Equation 1 using individual-level Spanish census microdata. Consistent with the baseline results presented in Table 1, Table B4 displays no detectable effect at the extensive margin, while a statistically significant effect appears when the intensive margin is also considered (i.e., the total number of children of a woman). However, in relation to the baseline analysis, two key differences stand out. First, the estimates are now statistically significant for both partially and fully treated cohorts. Second, the magnitude of the effect is considerably smaller in both absolute terms and relative to the mean of the control group. Quantitatively, women in partially treated cohorts have on average 0.04 fewer children compared to the control group, corresponding to approximately a 1% reduction relative to the average number of children among women born between 1920 and 1929. Women in fully treated cohorts have 0.08 fewer children on average, or about a 3% decrease relative to the mean of the control group.

The estimated coefficients obtained by estimating Equation 1 on the sample of Portuguese women from the ESS are presented in Table B5. Their inspection mitigates the concern that our findings may be merely driven by generational dynamics. Although Portugal is highly comparable to Spain under Franco - in terms of geographic proximity, strong Catholic tradition, and the presence of an authoritarian regime—women in the same birth cohorts who were partially or fully exposed to Franco's reform (in temporal terms), but born in Portugal show no significant changes in fertility outcomes, either at the extensive or intensive margin. This placebo result strengthens our conclusion that the effects observed in Spain are indeed attributable to exposure to the 1945 primary school reform, rather than to broader cohort-based trends affecting southern European countries during this period.

5.2 Mechanisms

Multiple mechanisms may be at work behind the effects documented in Table 1. First, a labor market substitution mechanism may be at play: women who were partially or fully treated by the reform may have increased their participation in the labor force, thus reducing the time available for childbearing, particularly when it comes to having more than one child. In contrast, the reform may have reduced the human capital of the treated cohorts by shifting the curriculum toward domesticity and motherhood-related instruction at the expense of skills that are directly valuable in the labor market. This reduction in labor market prospects may have, in turn, generated frustration or a sense of disempowerment, which could ultimately have resulted

in lower fertility.[26] In addition to these labor market channels, another plausible mechanism may operate through political identity. Being exposed to an authoritarian far-right regime that promoted traditional family values through education may have triggered a long-term ideological backlash among affected women, fostering more progressive or left-leaning attitudes that rejected the pro-natalist, conservative ideals associated with the regime. Finally, the reform may have directly altered both the primary and secondary beliefs of treated women about motherhood, gender roles, and the place of women in society, potentially affecting their attitudes towards childbearing more generally.

To explore the mechanisms that may drive our baseline findings, we reestimated the specification in Equation 1 using a set of alternative dependent variables: labor force participation, highest level of educational attainment, political orientation (measured as whether the respondent voted for right-wing parties in the most recent elections), and a battery of survey items capturing both primary and secondary beliefs related to motherhood. These belief-based outcomes include views on women's role within the family, attitudes toward children, participation in the labor market, and alignment with the teachings of the Catholic Church on fertility-related matters.

As reported in Table B6, we did not detect statistically significant differences between untreated, partially treated, and fully treated cohorts in their participation in the labor market, tertiary educational attainment, or voting behavior in favor of right-wing parties. These null results allow us to rule out most of the channels discussed above as primary drivers of our main findings.

In contrast, Figure 3 offers graphical evidence of the estimated coefficients using individual responses to survey questions on beliefs about women's roles in the family, the labor market, and motherhood in general. Interestingly, all estimated effects are directionally consistent with a mechanism by which exposure to the reform directly shaped treated women's values and beliefs, potentially inducing long-lasting changes in cultural attitudes. In particular, fully treated cohorts appear more likely to support gender equality in the domestic sphere, to reject the notion that men should do less housework, and to view equal task-sharing as essential for couple stability. These women also attribute lower importance to children in defining female identity or stability of relationships and place greater emphasis on women's participation in the labor market as a key component of family well-being. Finally, fully treated women are systematically less aligned with the traditional positions of the Catholic Church on fertility-related issues, including abortion, artificial insemination, and contraception.

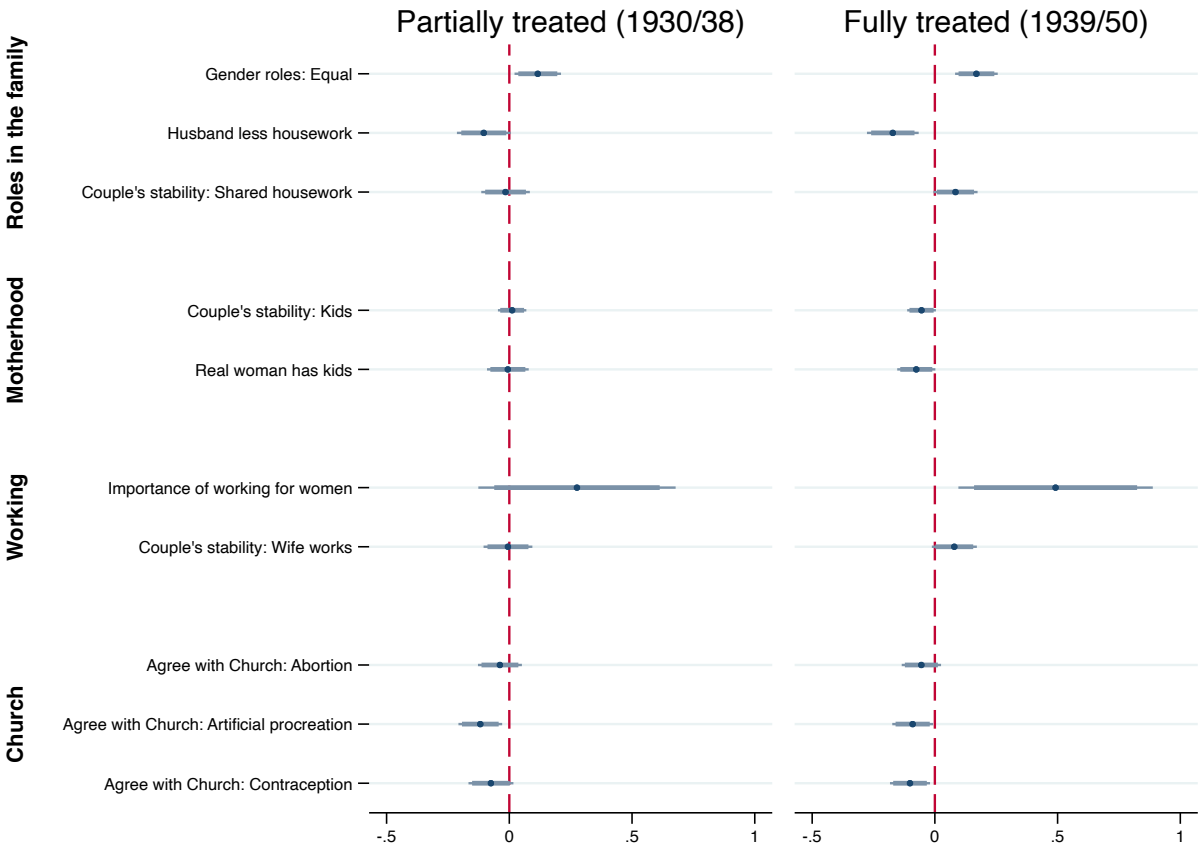
Partially treated cohorts exhibit similar patterns, although the magnitude of the estimated effects is generally smaller, and some coefficients fall short of conventional significance thresholds. While political preferences and social norms frequently move together, they are distinct and

[26] Grebol and Navarro (2025) show that Franco's regime disproportionately dismissed more competent teachers and that average experience decreased by approximately 2.5 years (14.2%), leaving persistent staffing deficits.

may respond differently to authoritarian legacies. Exposure to such regimes can reshape value systems and cultural attitudes without necessarily leading to partisan realignment.

Although these findings should be interpreted with caution given the limitations of these data, they offer at least strongly suggestive evidence to point towards a cultural backlash mechanism.[27] Specifically, our results are consistent with the hypothesis that exposure to the Francoist primary school reform, an intervention explicitly aimed at reinforcing traditional gender norms, may have induced a lasting rejection of the very same values it sought to establish, particularly among most exposed women.

Figure 3: The impact of Francoist primary school reform on personal beliefs



Note: The figure shows the point estimates and the confidence intervals at 90% (i.e. least-wide spikes) and at 95% (i.e. widest spikes) of the exposure to Franco's primary school reform on personal beliefs, for women born between 1930 and 1938 (left panel) and women born between 1939 and 1950 (right panel), both with respect to women born between 1920 and 1929. The dependent variables are women' beliefs on gender roles within the household, motherhood, women' participation to the labor force and alignment with Catholic values. All regressions include place of residence fixed effects and adjust standard errors for heteroskedasticity.

[27] In fact, belief-related variables are available only from a single survey wave (1990), which precludes the inclusion of year or age-class fixed effects in the corresponding regressions.

5.3 Analysis at the macro level

The final part of our empirical analysis investigates whether individual-level behavioral changes induced by the reform translated into detectable aggregate effects: specifically, whether fertility declined more sharply in localities where the population was more exposed to the policy. To test this, we adopted a difference-in-differences (DiD) approach at the aggregate level (i.e., at the locality level, defined as the provincial capitals and the rest of the provinces) and compared fertility trends before and after the 1945 primary school reform across localities with differential predetermined exposure to the reform. In particular, we contrasted locations with a higher pre-reform share of girls aged 0-14 in 1940, that is, the first cohorts to be (partially or fully) affected by the policy, with locations where this share was lower. Formally, we estimated the following equation:

$$y_{it} = \beta_0 + \beta_1 \, Treat_i \times Post_t + \mathbf{X}_{it} + \Gamma_i + \tau_{pt} + \epsilon_{it} \tag{3}$$

where i denotes a locality, i.e., an administrative unit defined either as a provincial capital or as the remainder of the province, and t indicates the year. The outcome variable y_{it} is the number of live births in unit i in year t. The interaction term $Treat_i \times Post_t$ combines $Post_t$, a dummy equal to 1 for years after 1945 (that is, after the reform), and $Treat_i$, which captures the share of the female population aged 0-14 in 1940 - corresponding to the cohorts that would have been of school age at the time of the reform. We also added a set of controls $\mathbf{X}_{it}$, which includes the interaction between $Post_t$ and the share of the male population aged 0–14 in 1940, as well as the number of women of reproductive age in each unit and decade, to ensure comparability of the raw birth counts over space and time, as discussed in Section 3.[28] The specification further includes unit fixed effects (Γ_i), which absorb time-invariant characteristics at the locality level, and province-by-year fixed effects (τ_{pt}), which account for time-varying shocks that may have differentially affected different Spanish provinces (e.g., the effect of the Spanish civil war). Standard errors are robust to heteroskedasticity and clustered at the level of the administrative unit (i). As with the baseline results, we also estimated an event study version of Equation 3 to assess the validity of the parallel trends' assumption and derive additional insight into the dynamics of the post-treatment effects.

Table B7 reports balancing tests between localities more and less exposed to the reform for a range of local characteristics, including proxies for religiosity, demographic composition of the resident population, political orientation of the electorate, and characteristics of the local educational system.[29] In virtually all cases, we find no statistically significant differences between

[28] Census counts of women aged 15–44, available every ten years, were converted into annual panel values, whereby the value from each census is assigned to all years within the corresponding decade.

[29] For these tests, we estimated cross-sectional regressions that mirror as closely as possible Equation 3,

localities differentially exposed to the reform.[30]

Table 2 reports the estimated coefficients from Equation 3. Columns 1 and 3 include fixed effects for year and administrative unit (i), as well as controls for the demographic composition of the fertile female population by age group and decade. Columns 2 and 4 further account for province-by-year fixed effects, thereby absorbing time-varying shocks at the provincial level. In addition, columns 3 and 4 incorporate an interaction term between the share of males aged 0-14 in 1940 and the post-treatment indicator ($Post$).

Across all specifications, the estimated coefficient for our main treatment variable - the interaction between the share of women aged 0-14 in 1940 and $Post$ - is negative and statistically significant.[31] The estimate in the fully saturated specification (column 4) indicates that a one-percentage-point increase in the share of the female population exposed to the primary school reform leads to a decrease of approximately 460 live births per year. This corresponds to roughly 9% of the average number of births recorded in statistical units over the five years preceding the introduction of the reform (i.e., 1939–1944). As a placebo, it is worth noting that the coefficient on the analogous interaction term for males (that is, the share of men aged 0-14 in 1940 $\times$ $Post$) is consistently much smaller, roughly one-third the size of the female coefficient, and never statistically significant. Thus, this pattern is consistent with the notion that the reform targeted girls directly and had limited spillover effects on male cohorts.

Based on the specification reported in column 4 of Table 2, Figure 4 presents the corresponding event study analysis. In Figure 4, we did not detect statistically significant differences in the number of live births between the treated and control areas in the years prior to the introduction of the reform, lending support to the assumption of parallel trends. In contrast, statistically significant and increasingly negative differences emerge shortly after the reform, growing in magnitude over time. This dynamic is mechanically driven by changes in the composition of the fertile female population: as time progresses, the cohorts exposed to the reform – either partially or fully – gradually replace older, unexposed cohorts, thereby increasing the share of women subject to the reform within the childbearing age range. The vertical red lines in Figure 4 help mark this transition. The lightest red line indicates the year when the youngest fertile age group (15-24) becomes almost entirely composed of partially exposed women; the medium

changing only the dependent variable. As these are cross-sectional regressions, we controlled for province fixed effects and for a dummy that distinguishes provincial capitals from the rest of the province.

[30]The only exception is the share of votes received by the Socialist party in the last free elections held in 1931, which is slightly lower in areas more exposed to Franco's educational reform. That is, these areas appear to historically have had weaker support for socialist parties. While the only requirement for the identification strategy is the validity of the parallel trends assumption, we nonetheless run robustness tests that allow for non-parametric differential trends based on pre-treatment socialist vote shares. These controls do not alter the results in terms of magnitude or significance.

[31]The effect becomes slightly noisier in Column 4 when both province-year fixed effects and the male-share interaction are included; the p-value in this case is 0.052.

Table 2: The impact of Francoist primary school reform on the number of births: DiD

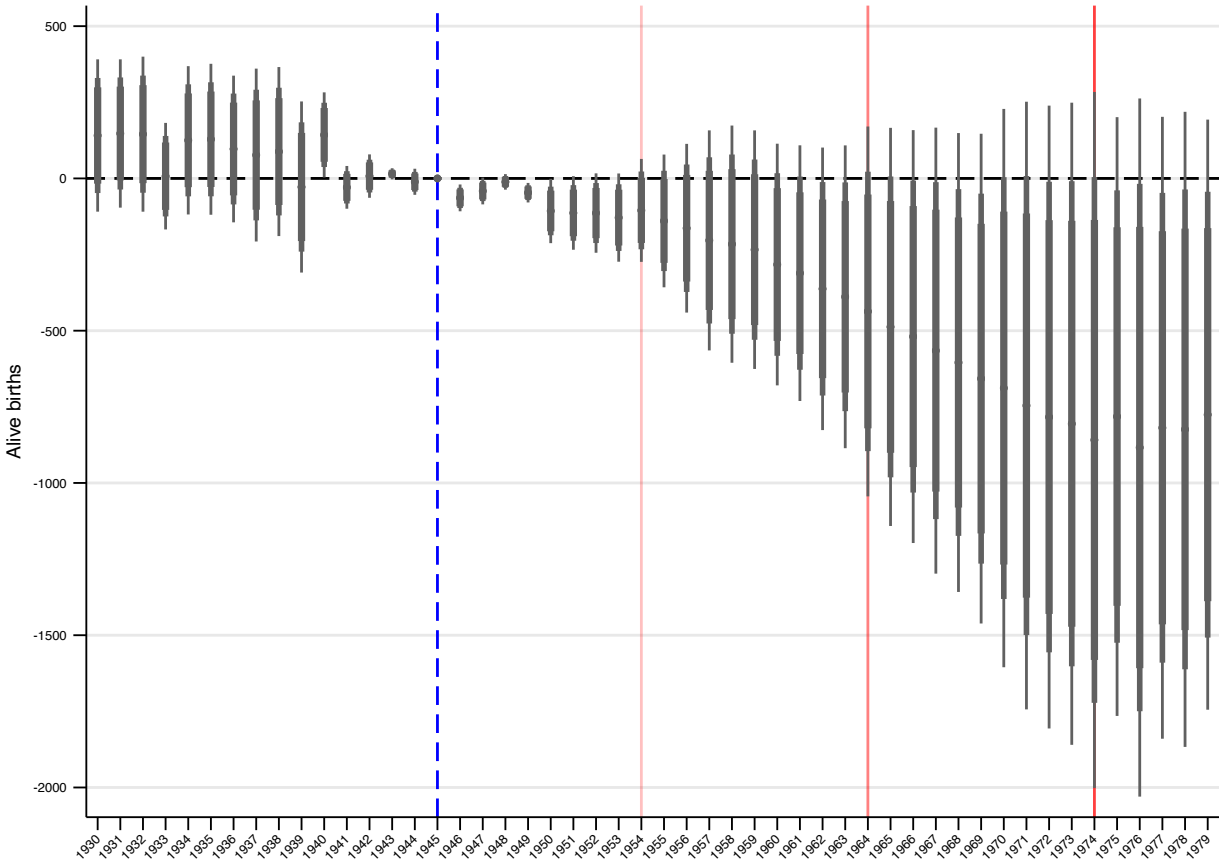
	(1) Births	(2) Births	(3) Births	(4) Births
Treat (W) X Post 1945	-429.396***	-519.120**	-392.784***	-459.933*
	(115.321)	(200.829)	(146.955)	(234.202)
Treat (M) X Post 1945			-55.138	-163.600
			(86.689)	(138.879)
Observations	4,919	4,918	4,919	4,918
R-squared	0.795	0.891	0.795	0.892
Unit FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Prov. X Year FE	NO	YES	NO	YES
Mean of depvar	5183	5183	5183	5183

Dependent variable: *Number of births*. The variable *Treat (W)* × *Post* represents the interaction between the share of women aged 0–14 in 1940 in each observational unit (*Treat (W)*) and the post-treatment indicator *Post*, which equals 1 for years after 1945 and 0 otherwise. Similarly, *Treat (M)* × *Post* is the interaction between the share of men aged 0–14 in 1940 (*Treat (M)*) and *Post*. All specifications include unit and year fixed effects, as well as controls for the size of the female reproductive-age population by age group and decade. Columns 2 and 4 additionally include province-year fixed effects. Standard errors are heteroskedasticity-robust and clustered at the unit level. The last row reports the average number of live births per unit in the pre-treatment period (1939–1944). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

red line corresponds to the point when the 15-24 group and the 25–34 group are composed entirely of partially or fully treated women; and finally, the darkest red line marks the year when the entire fertile population (ages 15-44) is composed exclusively of women either partially or fully exposed to the reform. Consistent with our expectations, the estimated treatment effect peaks around the latter threshold and stabilizes thereafter at a significantly negative level.

Table B8 presents the results of four different robustness checks. In columns 1 and 2, we address the potential imbalance in the share of votes for the Socialist Party in the 1931 elections—documented in Table B7—by including nonparametric trends based on this variable. Columns 3 and 4 replace the continuous treatment measure with a discretized version of this measure, since here we compare areas above and below the median of the exposure distribution. In columns 5 and 6, we perform a placebo test using as treatment the share of women aged 15-24 in 1940, a group who was too old to be directly exposed to the reform. Finally, columns 7 and 8 investigate whether the reform might have indirectly influenced birth rates through changes in marriage rates, given that marriage was at the time a near-prerequisite for childbearing decisions. All tests yield results that are consistent with our baseline findings and support the robustness of the main estimates presented in Table 2. Specifically, controlling for preexisting political alignment (i.e., Socialist vote share in 1931) has virtually no impact on the estimated effects. The discrete version of the treatment produces results that closely mirror those from the

Figure 4: The impact of Francoist primary school reform on fertility: Event Study



Note: The figure shows the point estimates and the confidence intervals at 90% (i.e. least-wide spikes) and at 95% (i.e. widest spikes) for each birth cohort of women of the event-study version of Equation 3 and comparable to the DiD estimates presented in column 4 of Table 2. The dependent variable is the number of live births in unit i at time t. The lightest red line indicates the year when the youngest fertile age group (15-24) becomes almost entirely composed of partially exposed women; the medium red line corresponds to the point when the 15-24 group and the 25–34 group are composed entirely of partially or fully treated women. The darkest red line marks the year when the entire fertile population (ages 15-44) is composed exclusively of women either partially or fully exposed to the reform.

continuous specification, both in sign and statistical significance. As expected and consistent with results using male exposure as placebo, replacing the treatment variable with the share of women aged 15-24 in 1940 yields a coefficient that is statistically indistinguishable from zero. The same holds when we examine marriage rates: areas with higher concentrations of reform-exposed women (aged 0-14 in 1940) do not exhibit significant differences in marriage outcomes relative to less-exposed areas.

Finally, one may be concerned that the estimates reported in Table 2 might merely reflect the broader process of the second demographic transition, a common phenomenon in industrialized countries. As discussed in Section 3, due to data limitations, we cannot construct suitable fertility measures for local units (e.g. Total Fertility Rate, TFR) that vary over time and allow spatial and temporal comparisons before and after Spain's industrialization. However, Figure B2 displays national TFR estimates by decade from 1850 to 2020, as reconstructed by the National Institute of Statistics of Spain (INE). Although Figure B2 confirms a long-term decline in fertility, it also shows that the TFR remained relatively stable during the post-treatment decades that are relevant for our analysis (that is, 1940-1970s). Thus, this provides suggestive evidence that our estimated decline in fertility was not driven by the demographic transition, but rather reflects the genuine effect of the Francoist primary education reform.

6 Conclusion

This paper examines the demographic consequences of the 1945 Francoist primary education reform in Spain, which sought to increase fertility by reinforcing traditional gender roles and promoting a nationalist-religious vision of women as wives and mothers. Exploiting quasi-exogenous variation in exposure across birth cohorts and localities, we show that the reform not only failed to achieve its intended goals, but in fact reduced fertility. Women who completed their primary education under the new curriculum bore significantly fewer children than comparable cohorts, with the effect operating primarily at the intensive margin. Complementary evidence suggests that this fertility decline was accompanied by a cultural backlash: treated women were more likely to reject the values promoted by the regime in terms of gender roles, family and motherhood. At the aggregate level, areas with a higher share of girls exposed to the reform subsequently experienced a marked reduction in fertility rates. Taken together, even though data limitations prevent us from isolating the contribution of each component of the reform, such as differences in curricula, class division, or teaching materials for boys and girls, these results demonstrate that attempts to engineer social norms through indoctrination in school may produce outcomes directly opposite to those envisioned by policymakers.

Our findings have important implications for contemporary policy debates. In a growing number of advanced economies, low fertility has ignited calls for a "return" to traditional values as a

path to demographic recovery. Public figures in the U.S. and several other Western democracies have advocated restoring more conservative family structures as a way to encourage childbearing. The evidence presented here challenges this narrative. Far from stimulating fertility, policies that seek to reassert gender stereotypes risk alienating those they target, thereby depressing rather than boosting demographic outcomes. In modern low-fertility settings, where women often bear the highest costs of childbirth, attempts to reinforce traditional gender roles may trigger resistance, especially when implemented through coercive policies in societies with limited gender equality or media pluralism.[32]

This study generates four lessons for pro-natalist policy. First, unlike economic incentives, stressing traditional gender roles through education is shown to be counterproductive, adding a new dimension to our understanding of policy effectiveness. Second, it provides suggestive evidence that state-led efforts to enshrine traditional roles can produce long-lasting changes in attitudes in the opposite direction. Third, it shows that authoritarian propaganda has the potential to extend beyond beliefs to core demographic behavior. Finally, our results highlight the limits of authoritarian attempts to redefine intimate life.

In conclusion, the Francoist reform illustrates the demographic risks of using education as a vehicle for ideological engineering. Rather than achieving a "demographic recovery", the policy provoked a cultural and demographic backlash with lasting effects. For modern policymakers confronting low fertility, this paper suggests that effective pronatalist strategies are not those that attempt to reimpose outdated gender norms.

[32]Orban's attempt to stimulate fertility in Hungary through conservative discourses and economic incentives arguably constitutes a recent example (Dunai and Romei, 2024).

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Appendix

A Data

Table A1: Summary Statistics

	Mean	Std. Dev.	Min	Max	N
Panel A: CIS Data					
Kids	0.902	0.298	0.00	1.00	4,157
N. kids	2.661	1.685	0.00	16.00	4,153
Gender roles: Equal	0.507	0.500	0.00	1.00	1,101
Husband less housework	0.545	0.498	0.00	1.00	903
Couple's stability: Shared housework	0.682	0.466	0.00	1.00	1,115
Couple's stability: Kids	0.842	0.364	0.00	1.00	1,111
Real woman has kids	0.639	0.480	0.00	1.00	1,081
Woman works	0.295	0.456	0.00	1.00	1,130
Couple's stability: Wife works	0.494	0.500	0.00	1.00	1,073
Agree with Church: Abortion	0.333	0.471	0.00	1.00	1,343
Agree with Church: Artificial procreation	0.267	0.443	0.00	1.00	1,204
Agree with Church: Contraception	0.266	0.442	0.00	1.00	1,198
Panel B: Census Microdata					
Kids	0.916	0.278	0.00	1.00	1,140,219
N. kids	2.649	1.690	0.00	20.00	1,140,219
Panel C: Macro level data					
Births	6001.111	7349.494	11.00	92270.00	4,919

Table A2: Questions from CIS

Survey *(Numb.)*	**Year**	**Question**
		Roles in the family
Intra-household inequality *(1867)*	1990	*Gender roles: equal.* As you know, there are currently different types of families. Of the three options I am about to read to you, could you tell me which one best matches your ideal family? ***a)*** A family in which both the man and the woman work outside the home and share household chores and childcare. *b)* A family in which the woman works fewer hours outside the home and therefore takes on a greater share of household chores and childcare. *c)* A family in which only the man works outside the home, and the woman is exclusively responsible for household chores and childcare.

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Table A2 – continued from previous page

Survey *(Number)*	**Year**	**Question**
Intra-household inequality *(1867)*	1990	*Husband less housework.* To what extent do you think a man should share household chores? *a)* Equally with the woman. *b)* To a lesser extent than the woman. *c)* Only in case of necessity.
Intra-household inequality *(1867)*	1990	*Couple's stability: Shared housework.* Now I am going to read you a series of attitudes and situations that some people believe contribute to a happy marriage. For each one, please tell me whether you think it is very important, quite important, slightly important, or not important at all for achieving the stability of the couple: sharing household chores.
		Motherhood
Intra-household inequality *(1867)*	1990	*Couple's stability: Kids.* Now I am going to read you a series of attitudes and situations that some people believe contribute to a happy marriage. For each one, please tell me whether you think it is very important, quite important, slightly important, or not important at all for achieving the stability of the couple: having children.
Intra-household inequality *(1867)*	1990	*Real woman has kids.* In the case of women, do you believe that being a mother is an experience one must have in order to truly feel like a woman? *a)* Yes, *b)* No, *c)* Don't know, *d)* No answer.
		Working
Questions about values *(2001)*	1992	*Importance of working for women* I am now going to read you a series of aspects of people's lives that are considered important. Please rate each one from 1 to 10, according to how important it is for you, knowing that 1 means it has no importance at all and 10 means it has the greatest importance.
Intra-household inequality *(1867)*	1990	*Couple's stability: Wife works.* Now I am going to read you a series of attitudes and situations that some people believe contribute to a happy marriage. For each one, please tell me whether you think it is very important, quite important, slightly important, or not important at all for achieving the stability of the couple: the woman works outside the home.
		Church
February Barometer *(1860)*	1990	*Agree with Church: abortion.* Do you tend to agree or disagree with the recommendations made by the Catholic Church regarding abortion?

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Table A2 – continued from previous page

Survey *(Number)*	**Year**	**Question**
February Barometer *(1860)*	1990	*Agree with Church: artificial procreation.* Do you tend to agree or disagree with the recommendations made by the Catholic Church regarding artificial fertilization methods?
February Barometer *(1860)*	1990	*Agree with Church: contraception.* Do you tend to agree or disagree with the recommendations made by the Catholic Church regarding contraceptives?

B Additional Tables and Figures

Table B3: The impact of Francoist educational reform on childbearing decision: Different clusters

	(1) Kids	(2) Kids	(3) N. kids	(4) N. kids
1930/1938	-0.006	-0.006	-0.177	-0.177
	(0.020)	(0.013)	(0.129)	(0.140)
1939/1950	0.002	0.002	-0.344**	-0.344**
	(0.026)	(0.013)	(0.146)	(0.088)
Observations	3,944	3,944	3,940	3,940
R-squared	0.175	0.175	0.209	0.209
Year FE	YES	YES	YES	YES
Mun. FE	YES	YES	YES	YES
Age class FE	YES	YES	YES	YES
Year X Age group X Residence FE	YES	YES	YES	YES
Clusters	Cohort	Two-way	Cohort	Two-way
Boot. P-val	.93	–	.048	–
Base value	.89	.89	2.84	2.84

Dependent variable: *Ever had a child* (Col. 1 and 2), *Number of children ever had* (Col. 3 and 4). All estimates are based on the 1979 and 1990–1992 waves of the CIS barometer surveys. *1930/1938* is a dummy equal to 1 for women born between 1930 and 1938; *1939/1950* is a dummy equal to 1 for women born between 1939 and 1950. The omitted category comprises women born between 1920 and 1929. All specifications include the full interaction between year, municipality of residence, and age group fixed effects. Standard errors are robust to heteroskedasticity and clustered at the birth cohort level (col. 1 and 3), or using a two-way clustering procedure both at the birth cohort and survey year level (col. 2 and 4). Columns 1 and 3 also report the p-value obtained using wild bootstrap clustering with 9999 repetitions. The last row reports the mean of the dependent variable for the reference category (i.e., women born between 1920 and 1929). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table B4: The impact of Francoist educational reform on childbearing decision: Census

	(1) Kids	(2) N.Kids
1930/1938	-0.005	-0.035***
	(0.008)	(0.012)
1939/1950	-0.006	-0.078***
	(0.009)	(0.017)
Observations	1,140,203	1,140,203
R-squared	0.008	0.027
Year FE	YES	YES
Residence FE	YES	YES
Age class FE	YES	YES
Base value	.91	2.78

Dependent variable: *Ever had a child* (Column 1), *Number of children ever had* (Column 2). All estimates are based on the 1991 and 2011 waves of the Spanish Census. *1930/1938* is a dummy equal to 1 for women born between 1930 and 1938; *1939/1950* is a dummy equal to 1 for those born between 1939 and 1950. The omitted reference category includes women born between 1920 and 1929. All specifications include the full interaction between year, municipality of residence, and age group fixed effects. Standard errors are robust to heteroskedasticity and clustered at the birth cohort × survey year level. The last row reports the mean of the dependent variable for the reference category (i.e., women born between 1920 and 1929). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table B5: The impact of Francoist educational reform on childbearing decision: Portugal

	(1) Kids	(2) N. kids
1930/1938	-0.018	-0.629
	(0.020)	(0.399)
1939/1950	-0.061	-0.702
	(0.039)	(0.427)
Observations	751	751
R-squared	0.015	0.025
Year FE	YES	YES
Age class FE	YES	YES
Base value	.88	2.89

Dependent variable: *Ever had a child* (Column 1), *Number of children ever had* (Column 2). All estimates are based on the 2006 and 2018 waves of the European Social Survey (ESS) for Portugal. *1930/1938* is a dummy equal to 1 for women born between 1930 and 1938; *1939/1950* is a dummy equal to 1 for those born between 1939 and 1950. The omitted reference category comprises women born between 1920 and 1929. All specifications include year-age group fixed effects. Standard errors are robust to heteroskedasticity and clustered at the birth cohort × survey year level. The final row reports the mean of the dependent variable for the reference group (i.e., women born 1920–1929). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table B6: The impact of Francoist educational reform on labor force participation, education and political ideology

	(1) Employed	(2) Unemployed	(3) Housewife	(4) Educated	(5) Right-wing
1930/1938	0.017	0.006	-0.024	-0.004	-0.069
	(0.052)	(0.011)	(0.050)	(0.015)	(0.057)
1939/1950	0.075	-0.001	-0.074	0.002	0.032
	(0.058)	(0.018)	(0.058)	(0.020)	(0.097)
Observations	3,241	3,241	3,241	3,490	1,033
R-squared	0.253	0.190	0.257	0.162	0.327
Year FE	YES	YES	YES	YES	YES
Residence FE	YES	YES	YES	YES	YES
Age class FE	YES	YES	YES	YES	YES
Base value	.27	.03	.69	.02	.3

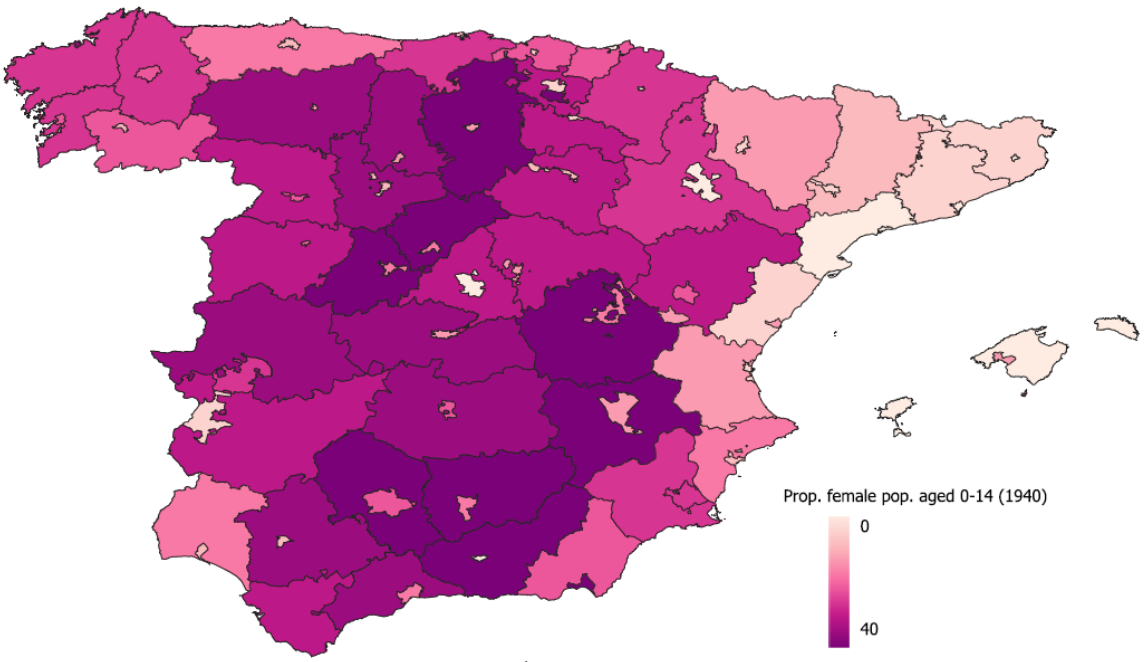
Dependent variables: *Employed (col. 1), Unemployed (col. 2), Housewife (col. 3), Tertiary Education (col. 4), Voted for Right-Wing Parties (col. 5).* All results except column 5 rely on the 1979 and 1986–1988 waves of the CIS Barometer; column 5 uses only the 1986–1988 waves. *1930/1938* is a dummy equal to 1 for women born between 1930 and 1938; *1939/1950* is a dummy for women born between 1939 and 1950. The reference group consists of women born between 1920 and 1929. All specifications include the full set of fixed effects defined by the interaction between year, place of residence, and age group. Standard errors are robust to heteroskedasticity and clustered at the year-by-year-of-birth level. The last row reports the average value of the dependent variable for the reference group (i.e., women born between 1920 and 1929). *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table B7: Treated vs Control - Balancing Tests

	(1) Coefficient	(2) SD
Purged teachers	0.328	0.818
Religious communities (1930)	-0.006	0.008
Professed (pc, 1930)	-0.175	0.125
Novices (pc,1930)	-0.020	0.051
Lay (pc, 1930)	-0.020	0.028
Republicans (%, local 1931)	0.465	0.487
Socialists (%, local 1931)	-0.567**	0.274
Monarchic (%, local 1931)	0.015	0.537
Comunists (%, local 1931)	0.014	0.031
Pro regime (%, 1947)	0.242	0.410
National schools (1000s,1942)	0.019	0.031
Primary teachers (1000s,1942)	0.012	0.030
Enrolled students (1000s,1942)	-0.040	1.239
Area in the war front (%)	0.234	0.738
Women aged 15-24 (%, 1940)	0.000	0.002
Women aged 25-34 (%, 1940)	0.000	0.001
Women aged 35-44 (%, 1940)	-0.000	0.001
Women aged 15-24 (1940)	-781	608
Women aged 25-34 (1940)	-764	553
Women aged 35-44 (1940)	-668	415
Male deaths (1000s, 1936-39)	0.289	0.714
Female deaths (1000s, 1936-39)	0.029	0.409

Note: This table presents balancing tests between treated and control areas across various local characteristics. Each covariate is regressed on the treatment indicator $Treat_i$. All specifications include province fixed effects and a dummy variable distinguishing provincial capitals to the rest of the province.

Figure B1: Proportion of female population aged 0-14 in 1940



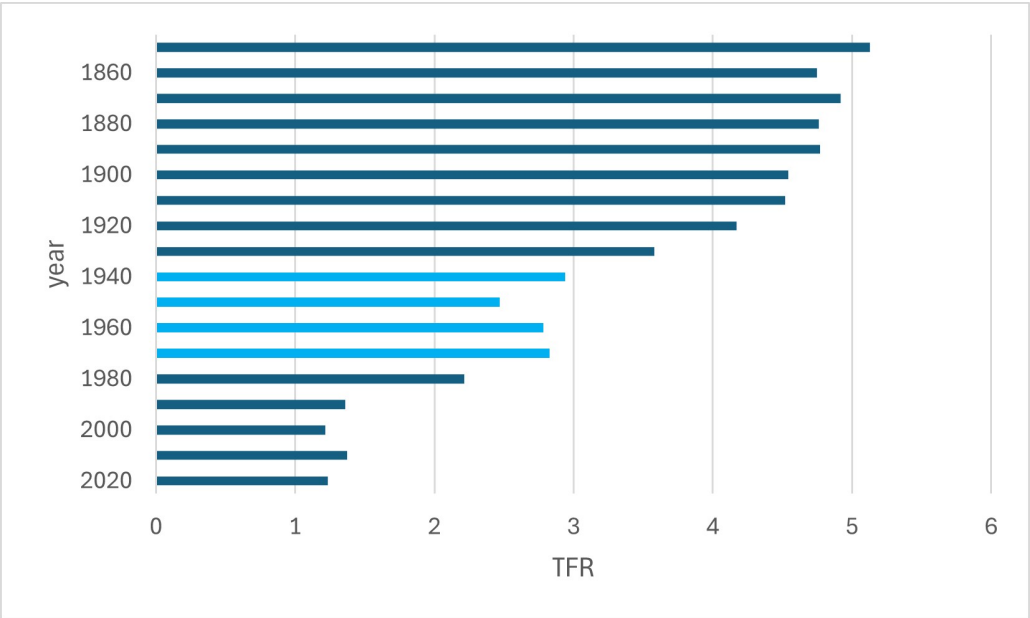
Note: The map shows the proportion of the female population aged 0–14 in 1940. Darker shades indicate a higher proportion of young females relative to the total female population.

Table B8: The impact of Francoist educational reform on the number of births: DiD - Robustness

	(1) Births	(2) Births	(3) Births	(4) Births	(5) Births	(6) Births	(7) Weddings	(8) Weddings
Treat (W) X Post 1945	-384.141***	-444.410**	-4,915.460***	-5,103.000**	122.977	107.961	-40.537	-62.790
	(138.237)	(214.473)	(1,701.797)	(2,417.523)	(115.894)	(111.838)	(55.894)	(65.155)
Treat (M) X Post 1945		-167.086		324.529		42.671		61.509
		(169.578)		(1,456.580)		(49.064)		(41.525)
Observations	4,919	4,918	4,918	4,918	4,918	4,918	4,918	4,918
R-squared	0.795	0.903	0.891	0.891	0.891	0.891	0.911	0.912
Test	Socialists	Socialists	Discrete	Discrete	Placebo	Placebo	Weddings	Weddings
Fertile pop.	YES	YES	YES	YES	YES	YES	YES	YES
Unit FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Prov. X Year FE	YES	YES	YES	YES	YES	YES	YES	YES
Mean of depvar	5183	5183	5183	5183	5183	5183	1728	1728

Dependent variable: *Number of births (cols. 1–6); Number of weddings (cols. 7–8).* In columns 1–2 and 7–8, *Treat (w)* × *Post* denotes the interaction between the share of women aged 0–14 in 1940 in each observational unit (*Treat (w)*) and the post-treatment indicator *Post*, which equals 1 for years after 1945 and 0 otherwise. In columns 3–4, *Treat (w)* is a dummy equal to 1 if the unit belongs to the upper half of the distribution of the share of women aged 0–14 in 1940, and 0 otherwise. In columns 5–6, *Treat (w)* refers to the share of women aged 15–24 in 1940. All specifications include unit, year, and province-year fixed effects, as well as controls for the size of the female reproductive-age population by age group and decade. Columns 1–2 also control for the interaction between year fixed effects and the share of votes received by the Socialist party in 1931 at the unit level. Standard errors are robust to heteroskedasticity and clustered at the unit level. The last row reports the average number of births per unit during the pre-treatment period (1939–1944). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure B2: Total Fertility Rate in Spain



Note: The figure displays the Total Fertility Rate (TFR) in Spain from 1860 to 2020. TFR represents the average number of children a woman would have over her lifetime based on age-specific fertility rates observed in a given year. Sources: Instituto Nacional de Estadística (INE), United Nations Population Division, and Statista.